

POC Slide

AI FLOOD PREDICTION APPLICATION

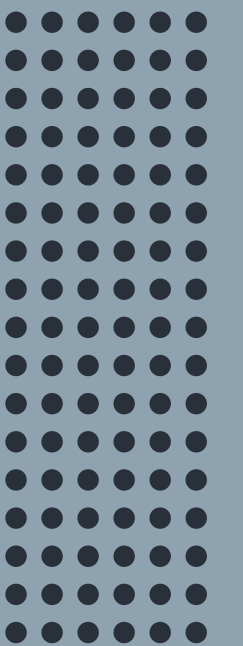
 **Group 7**

CHUA JIA LIN A23CS0069

NAZMI HAIKAL BIN KHAIRUL A23CS0145

POH LOK YEE A23CS0262

TAN ZHI MING A23CS0189



Introduction

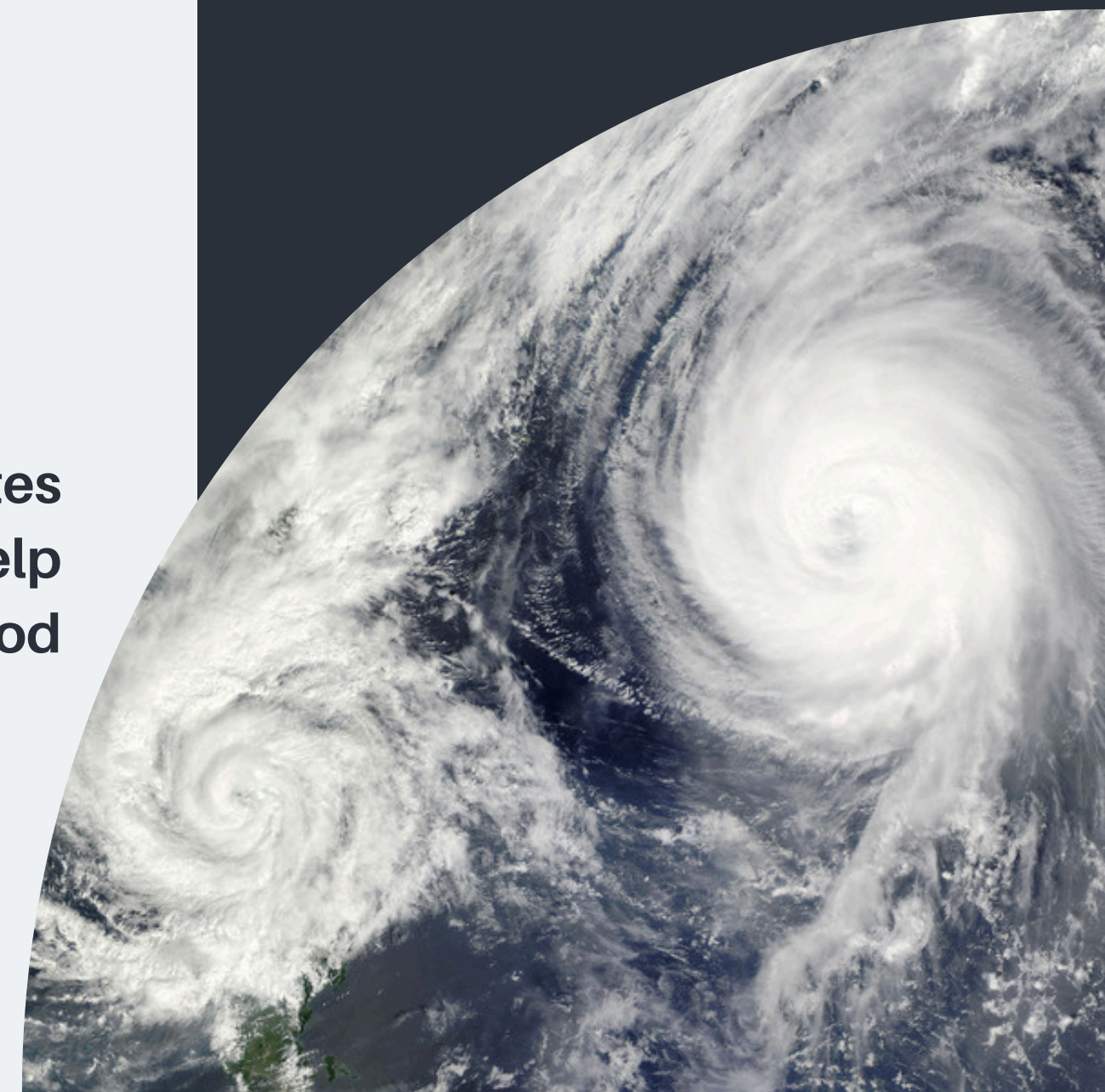
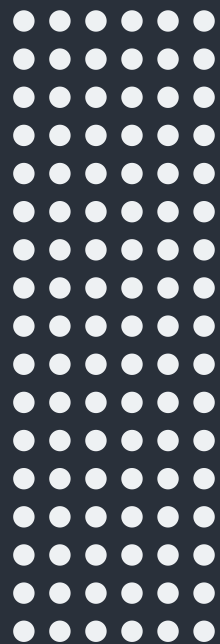
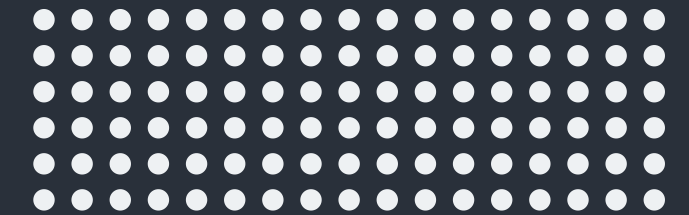


The AI Flood Prediction App is a smart city solution designed to monitor environmental conditions and predict flood risks in real time.

It uses IoT sensor data such as

- rainfall
- river level
- humidity
- drainage conditions
- past floods patterns

to provide early flood warnings. This prototype demonstrates how real-time monitoring and predictive analysis can help residents and authorities respond quickly and reduce flood impact.



Login Page

**User Login by
Several Ways**

 Continue with Email

 Login with Facebook

 Sign in with Apple ID

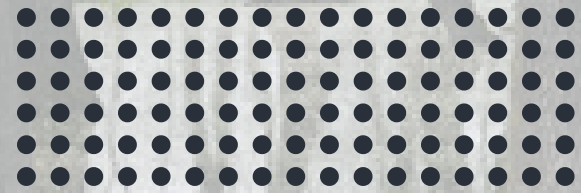
OR

 Continue as Guest

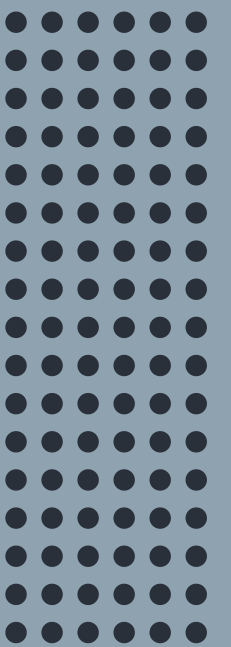
 Maintenance Team Access

By logging in, you agree to our Terms of Service and Privacy Policy.

**Maintenance Team
Login**



User Interface



A decorative graphic consisting of a grid of blue dots, arranged in 5 rows and 10 columns, totaling 50 dots.

Access to Notification Page

Critical Notification Display

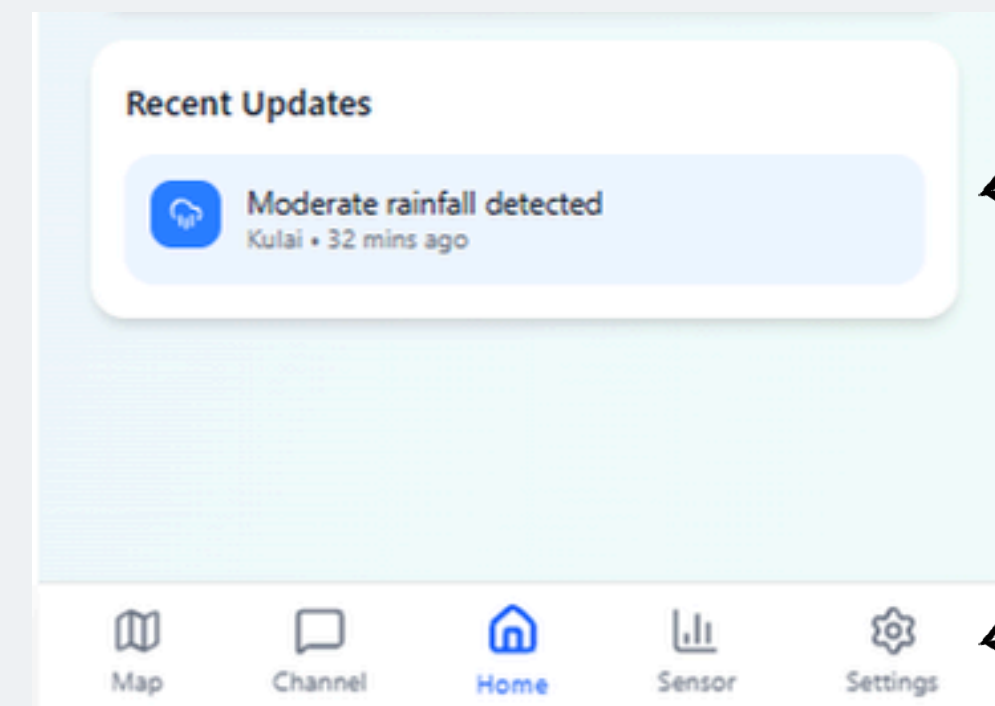
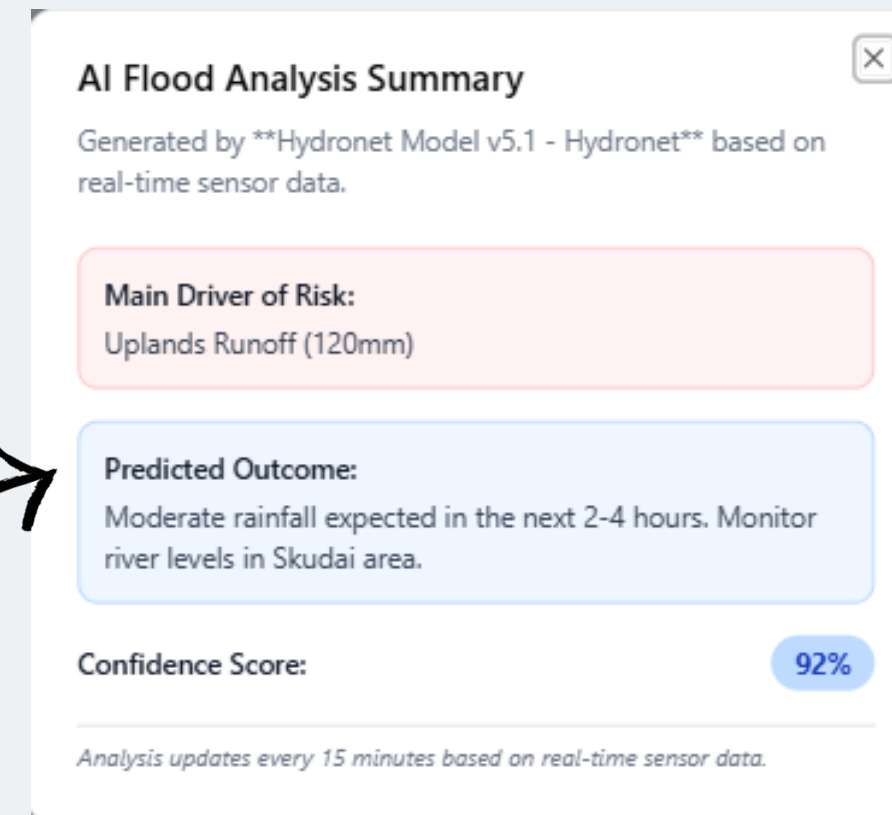
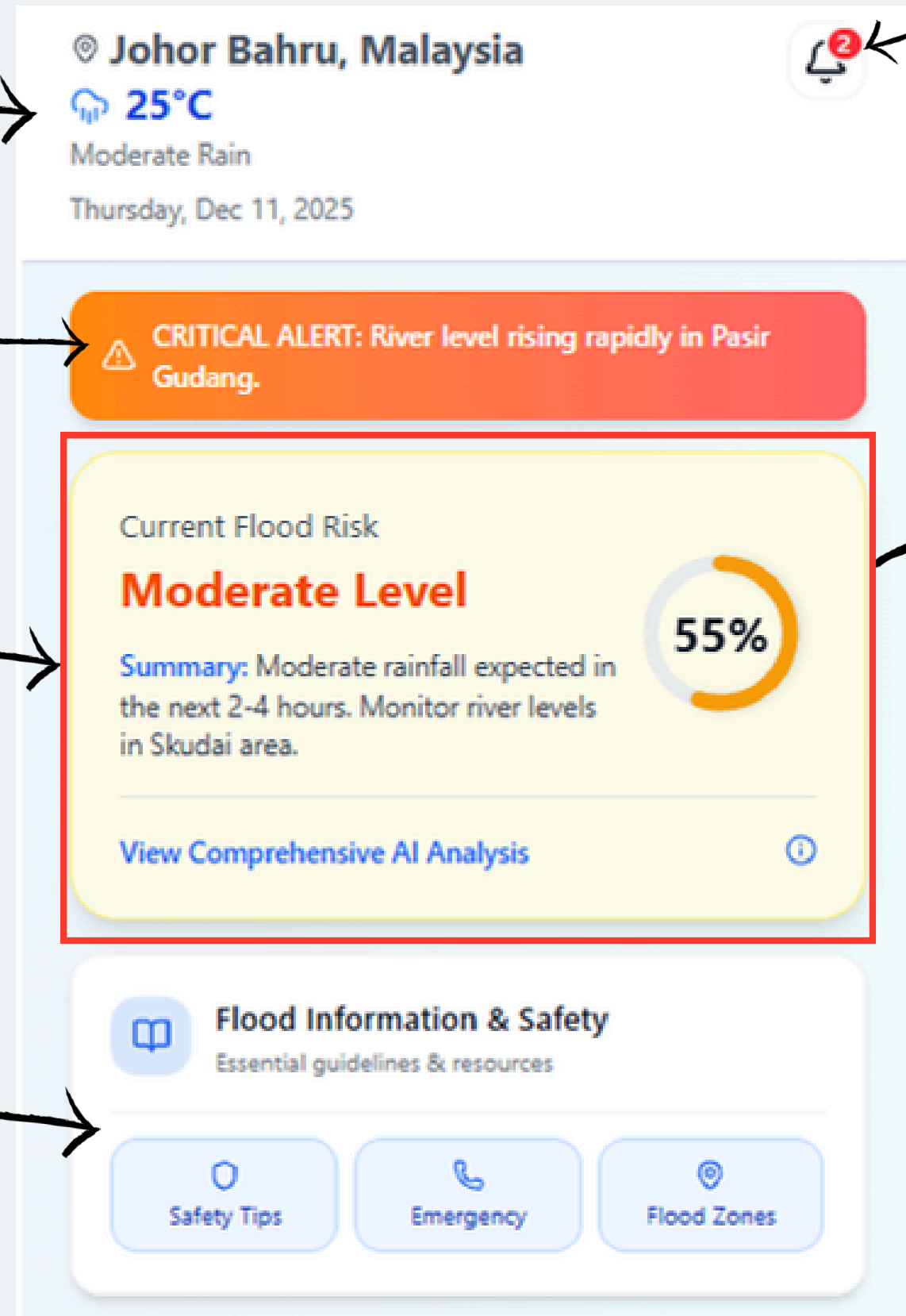
AI Prediction Of User's current area Flood Risk

Flood Information & Safety Guidelines

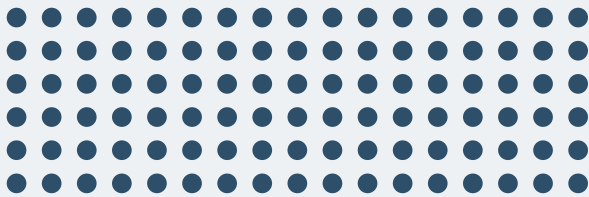
AI Flood Analysis Summary

Recent Updates Section

Navigation Bar

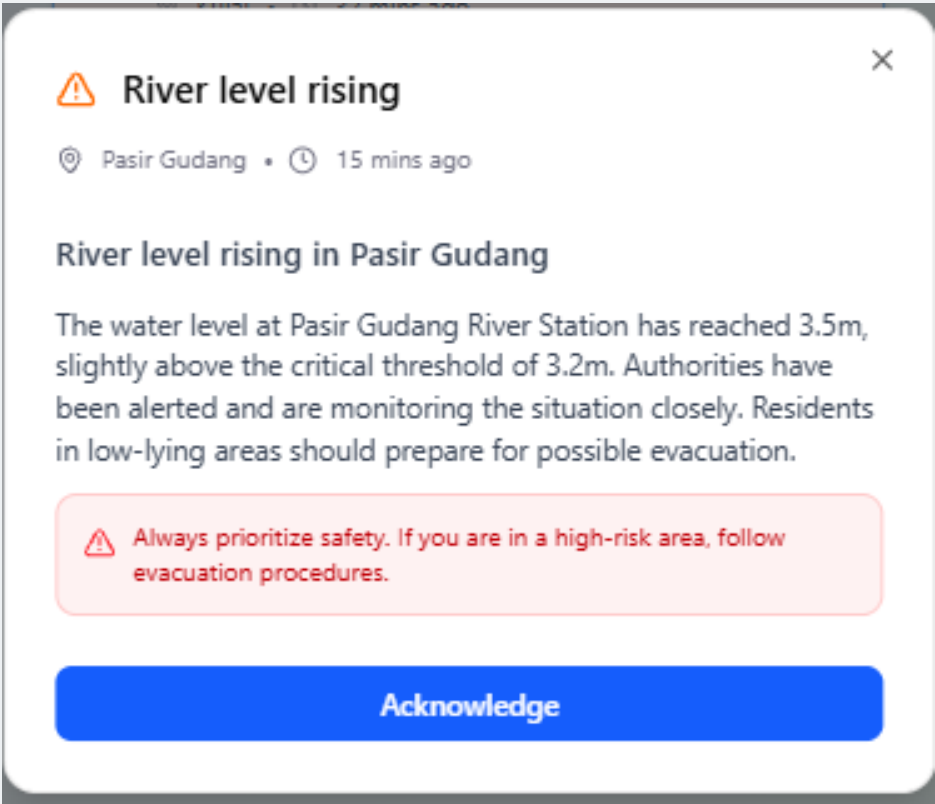
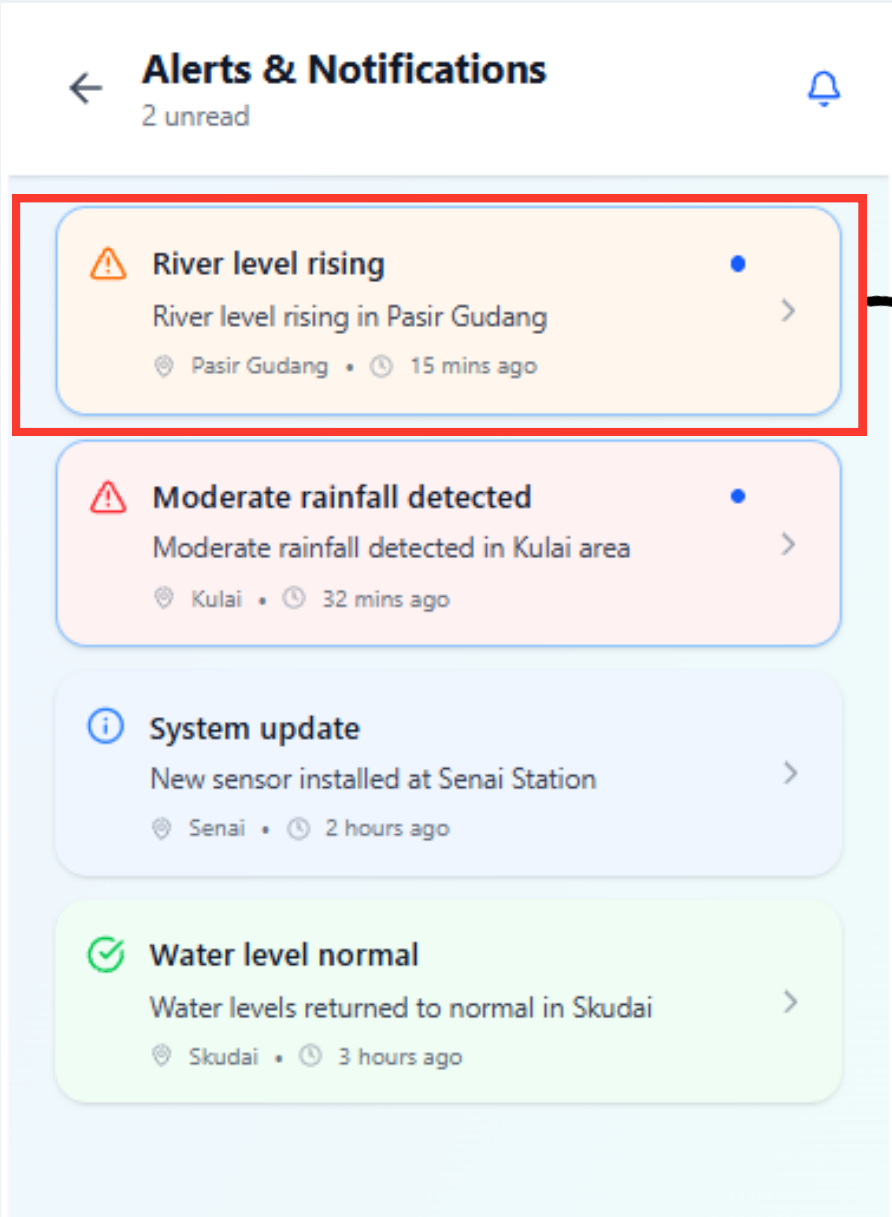


Notification Page



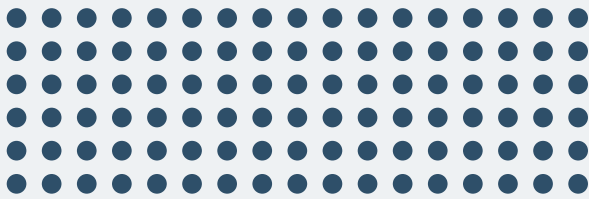
Notification and Alert
Received when Risk Increase

Notification List,
blue dot as
unread message



Notification
Details

Live Map Page



Flood Risk Real-time Mapping

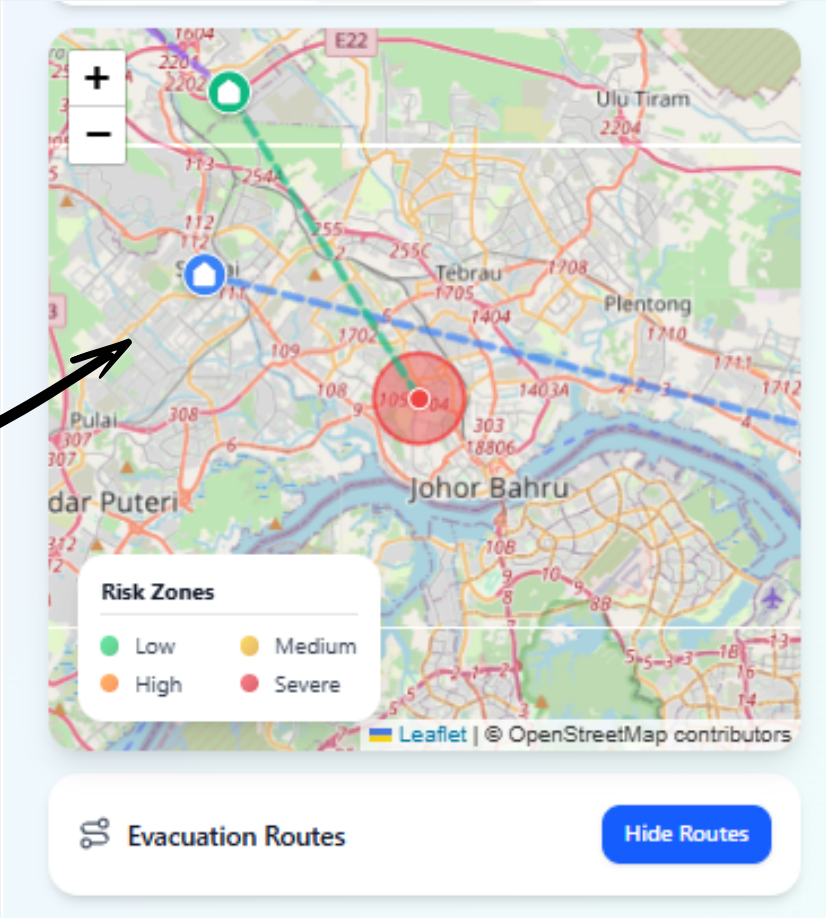
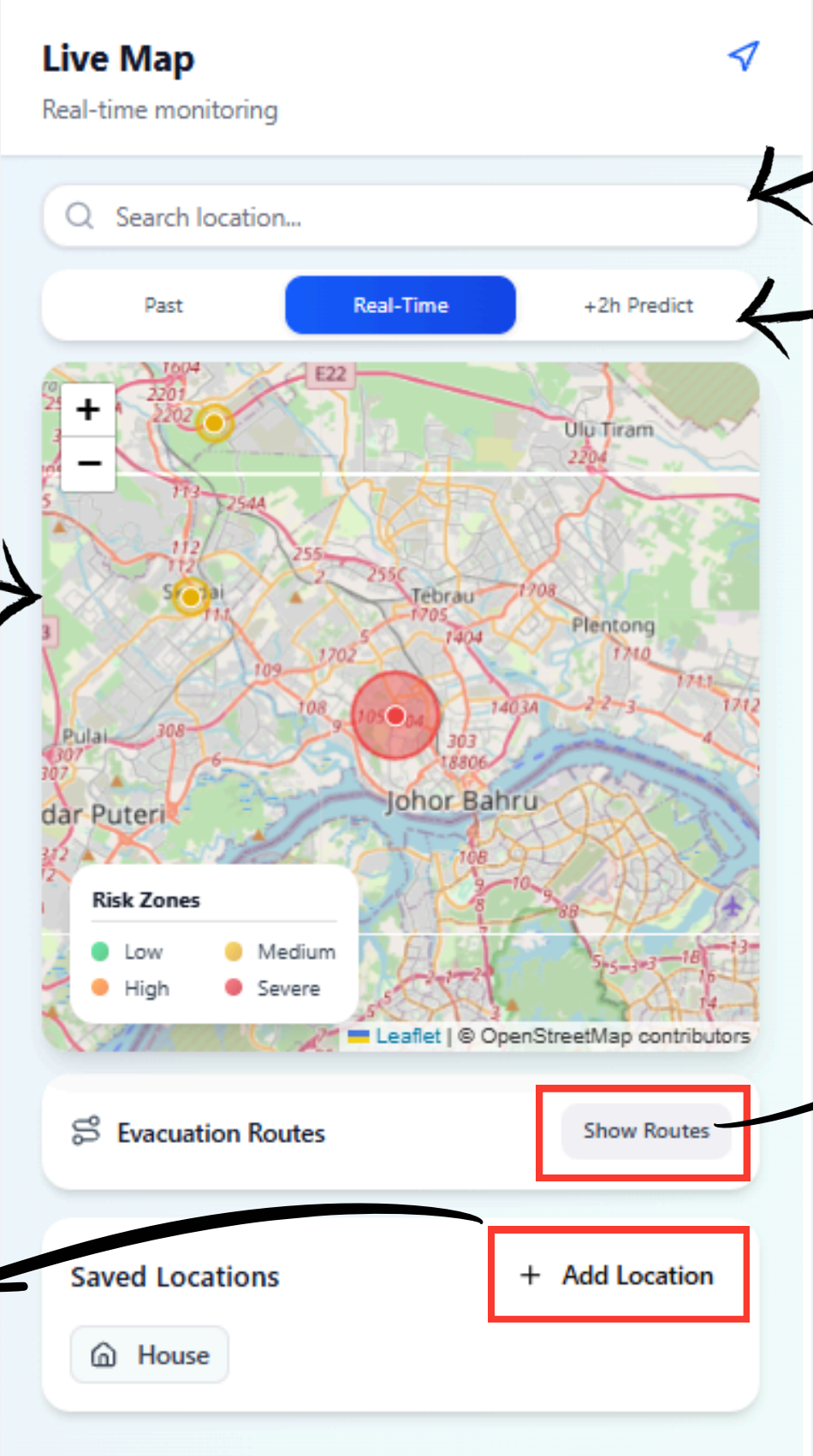
Location Changes

Past, Real-time and Future Flood Risk Prediction Viewing

Real-time Map with Risk Stated by Zone

Saved Location Function - Add Pin Point Location at Map

Evacuation Route shown for user's guide



Saved Locations

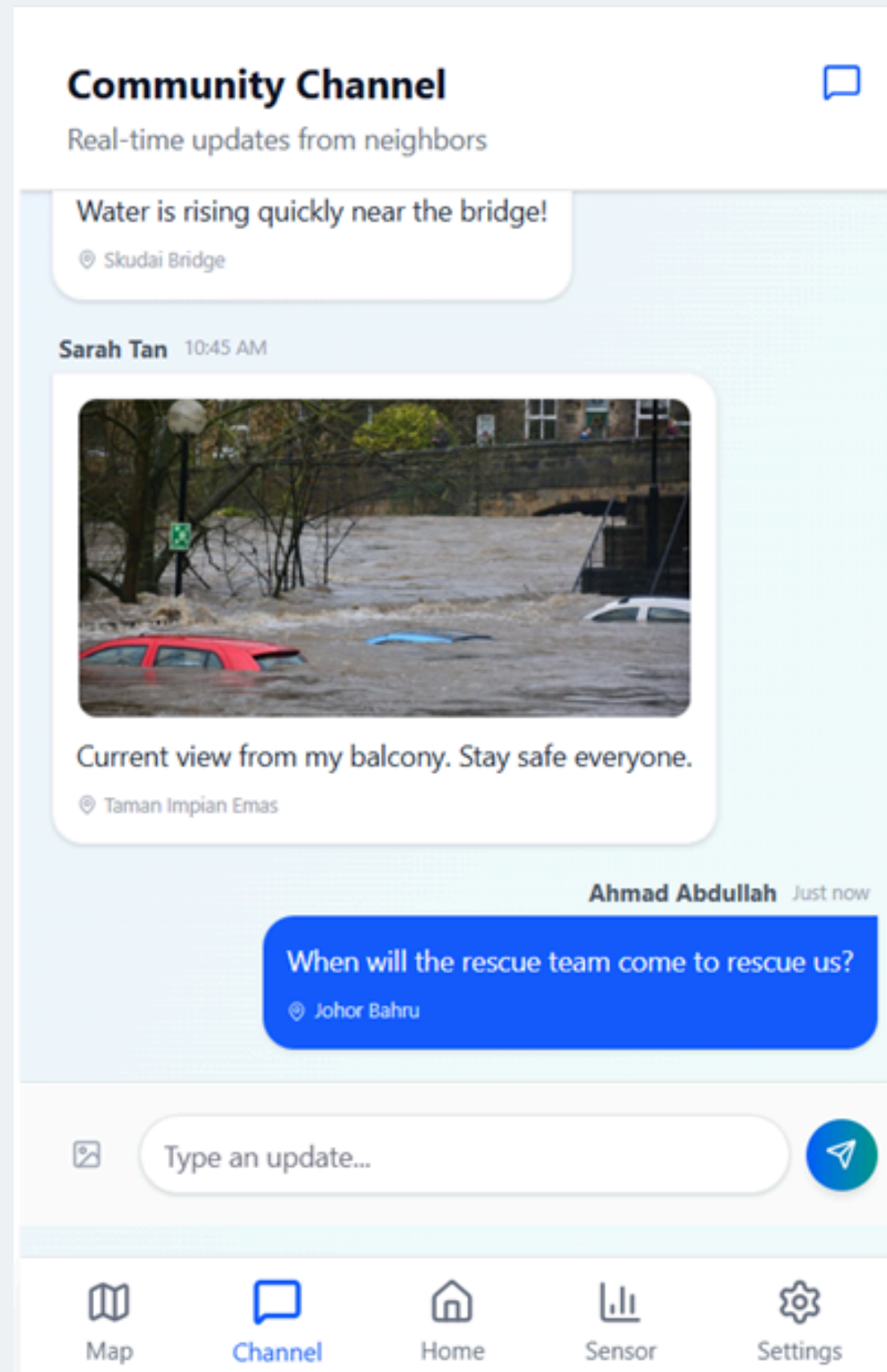
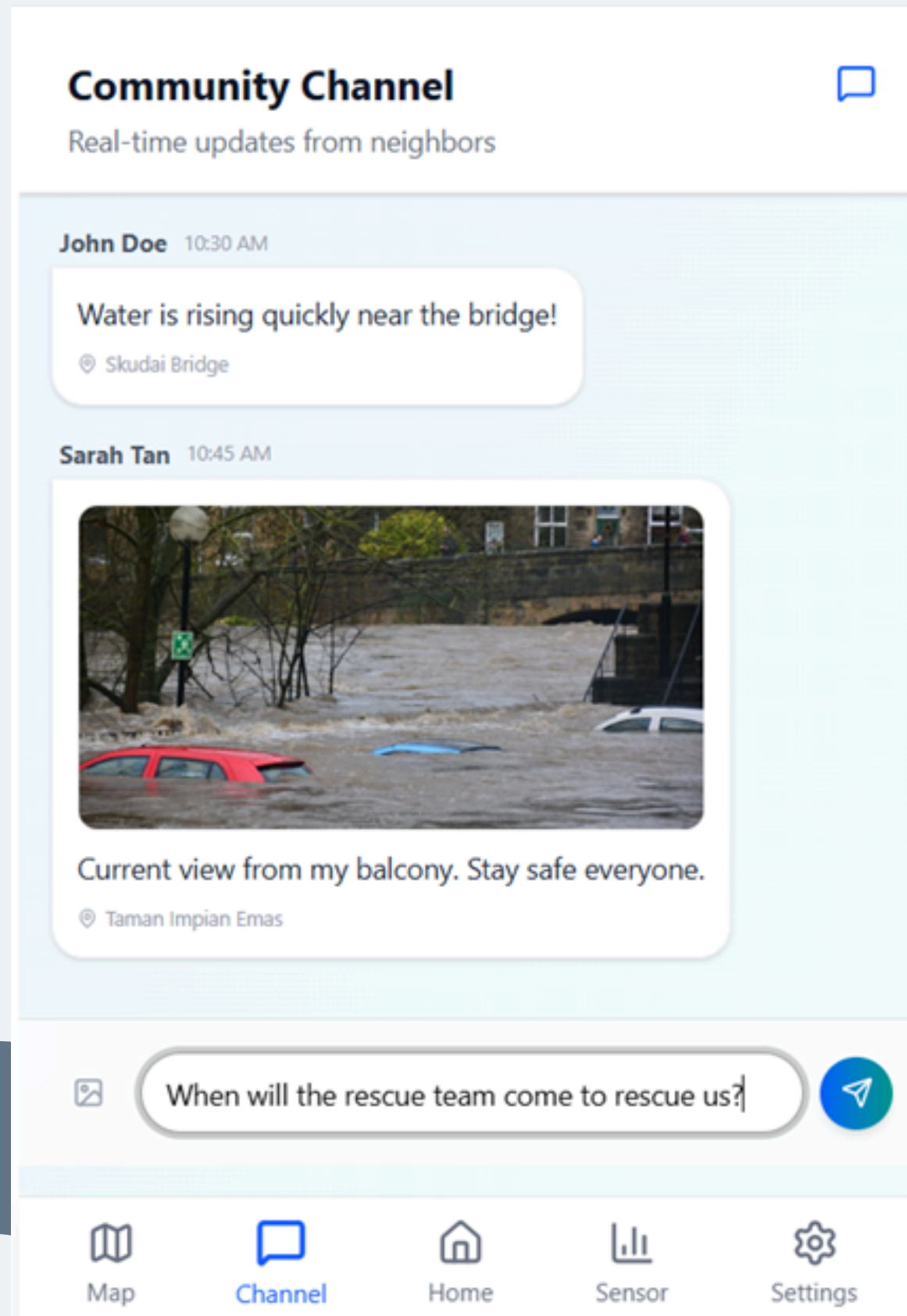
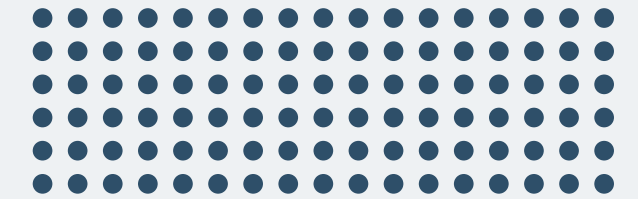
New Location Details

School

SMKSK

Saved Locations

Channel Interface

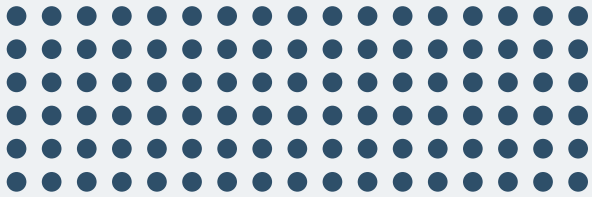


**Real-time
Communication
Channel**

< share latest information

< support text and image

Sensor Interface

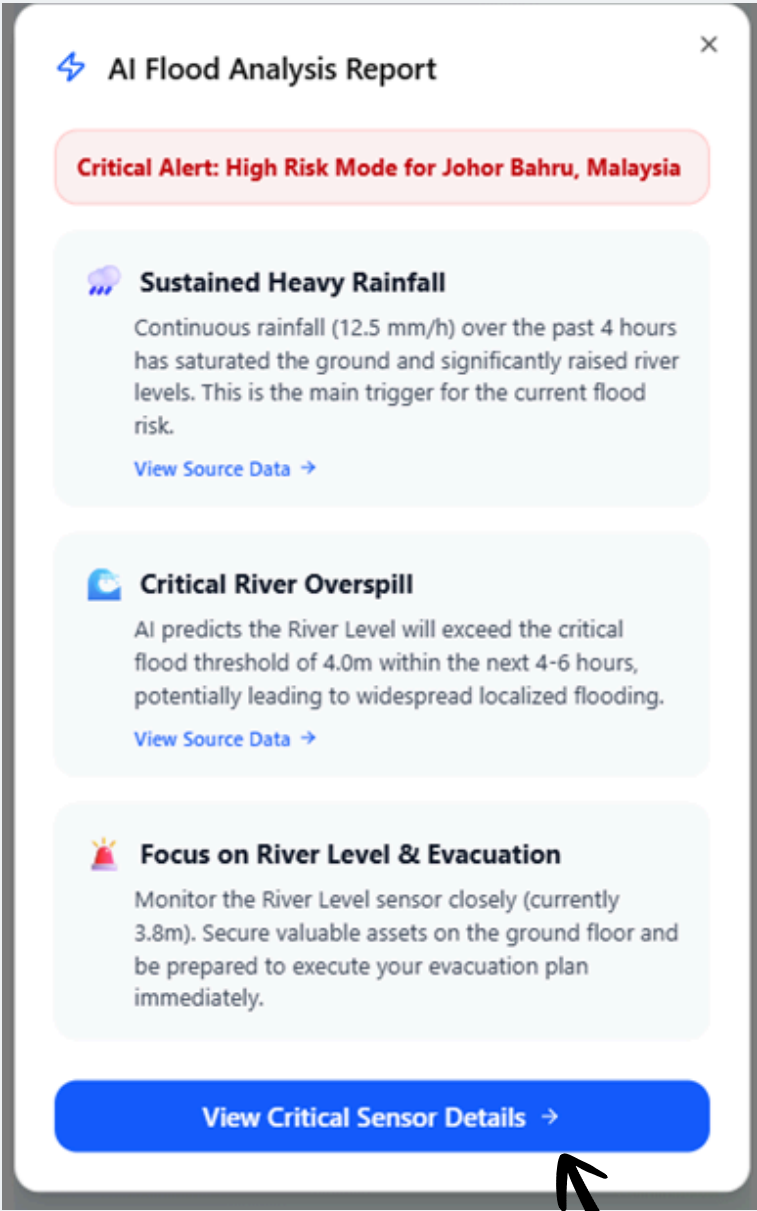
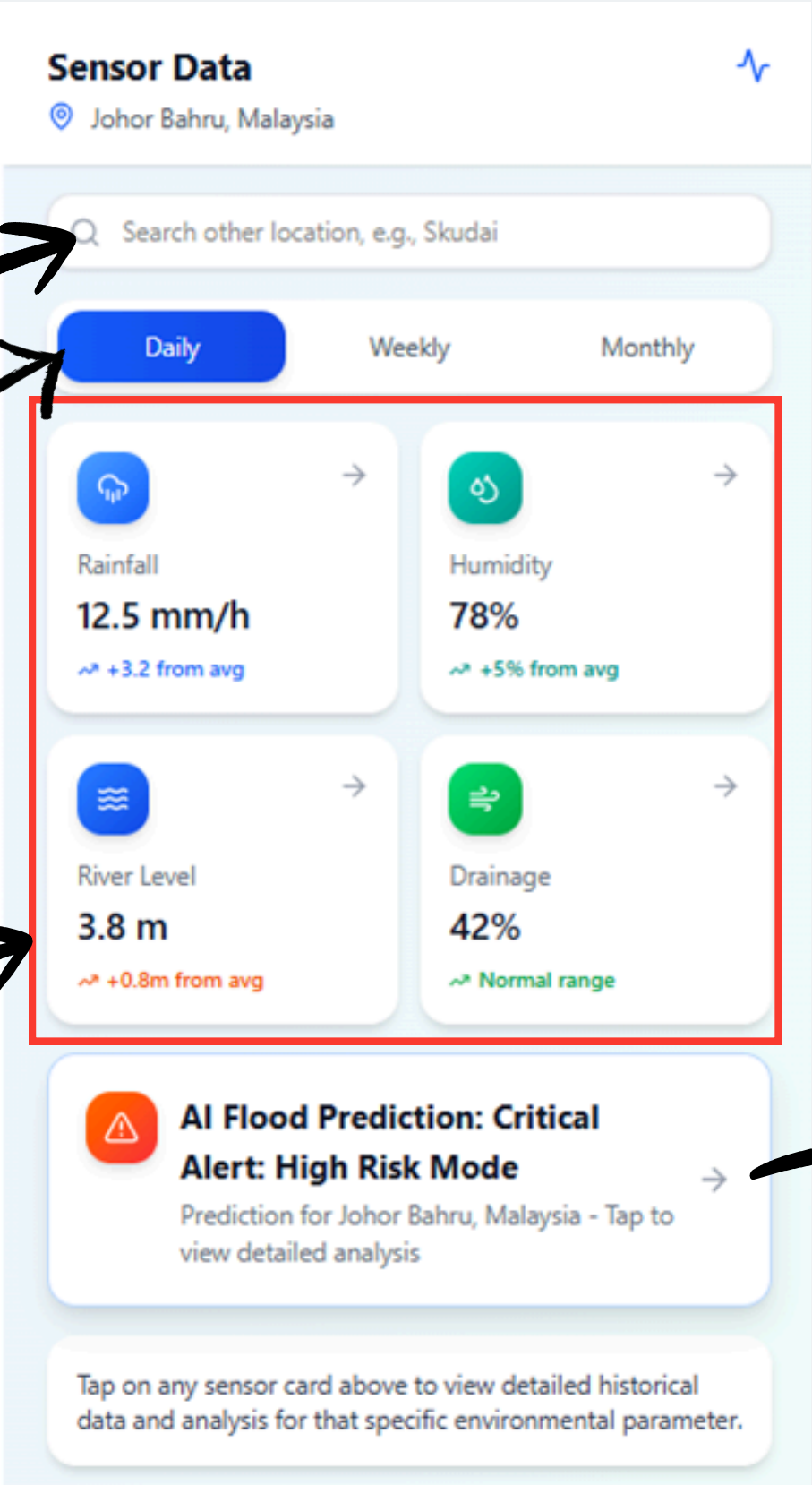


Real Time Environment
Data of Specific Location

Location Changes

View Short-term or
Long-term
Environmental trends

Key Flood Related
Parameters Display

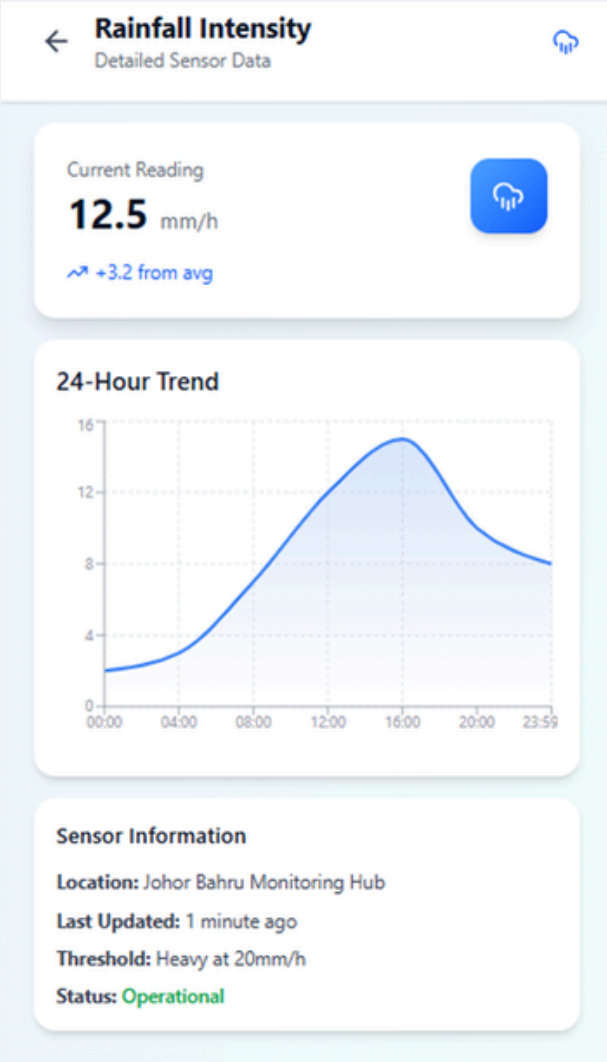


AI Model
Prediction
Report Based on
Environmental
Data

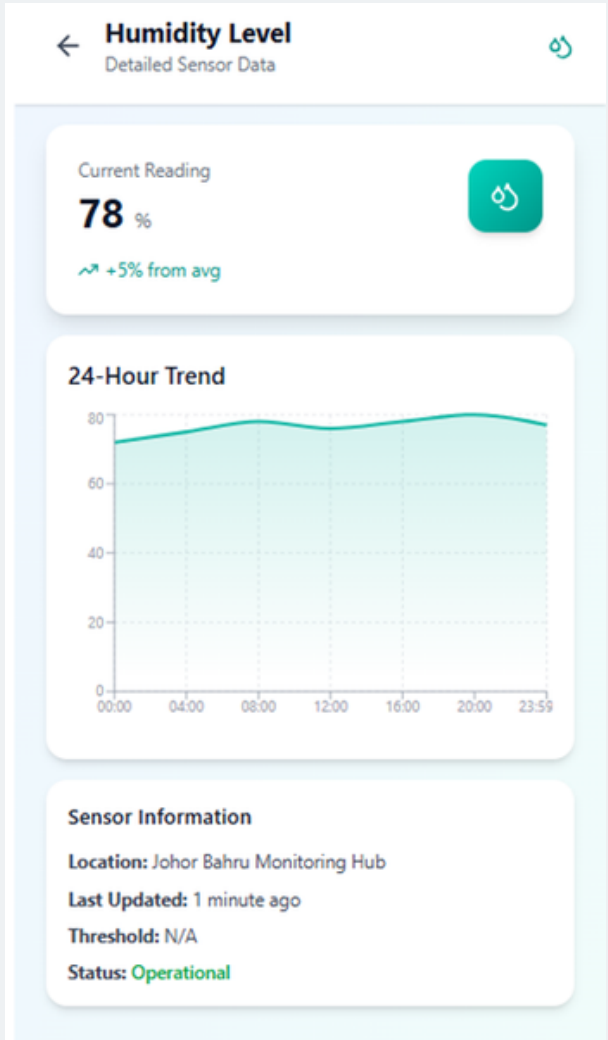
will navigate to critical
sensor parameter details

Sensor Details Interface

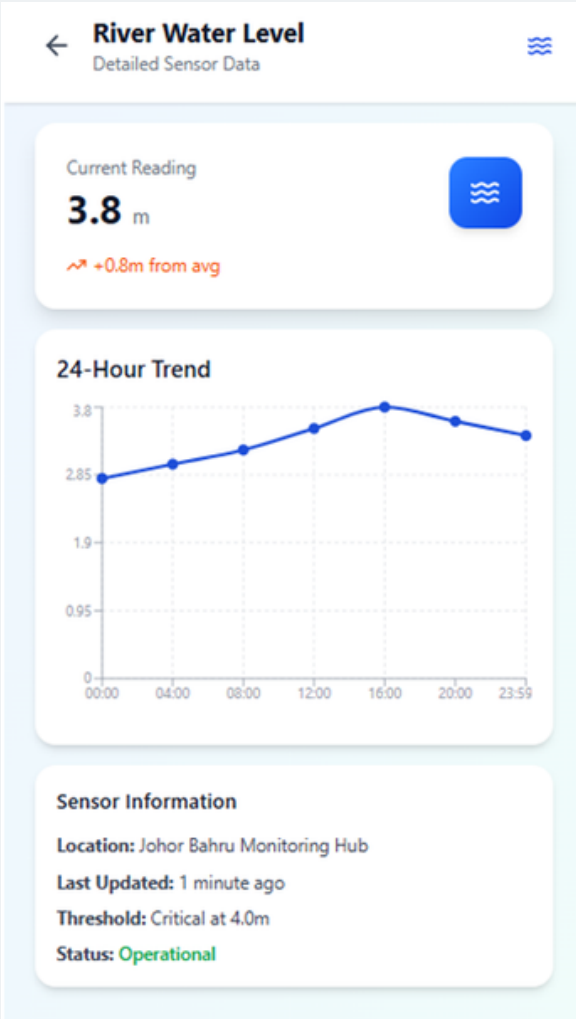
Specific Environmental Data can be Accessed to View Detials



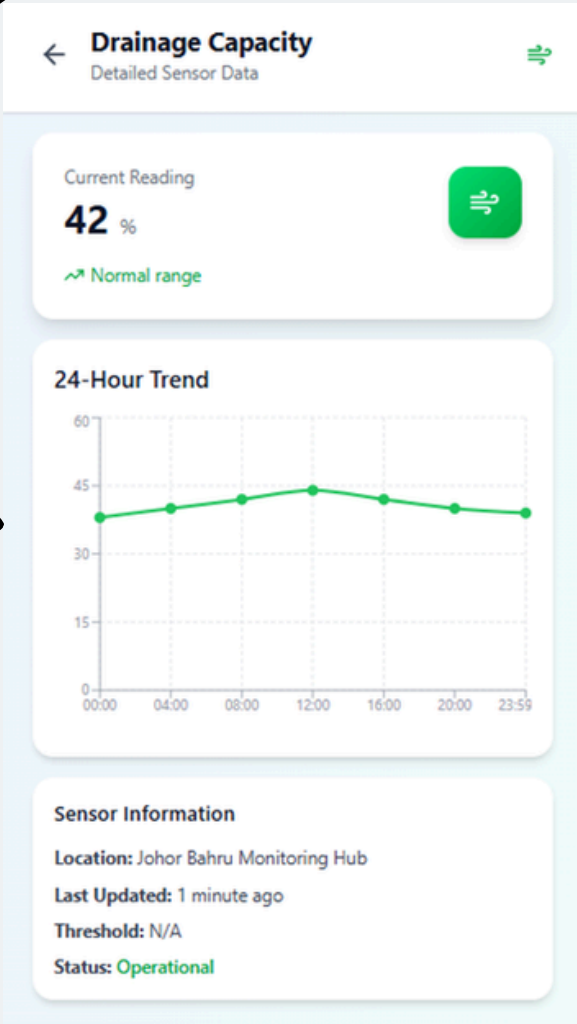
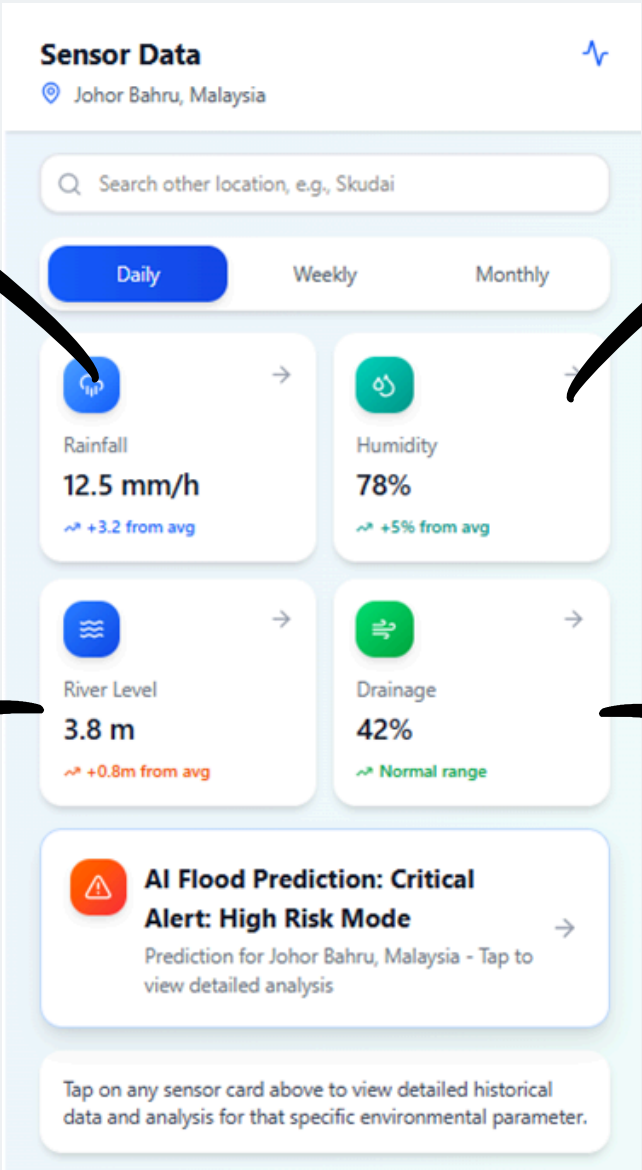
Rainfall Intensity Data



Humidity Level Data

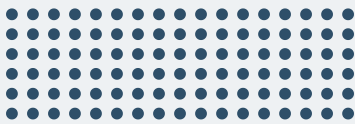


River Water Level Data



Drainage Capacity Data

Setting Interface



Personalize Settings

Notification Setting
to customize alert
message

Settings

Manage your preferences

Ahmad Abdullah

ahmad.abdullah@email.my

Alert Preferences

Push Notifications

Instant alerts on your device

SMS Alerts

Critical warnings via text

Sound & Vibration

Alert tones for warnings

Alert Severity

Choose which types to receive

Critical Alerts

Important Alerts

Info & Updates

Location
Setting

Location

Primary Location

Johor Bahru, Malaysia

Auto-detect Location

Use GPS for alerts

Language

English / Bahasa

Privacy & Security

Help & Support

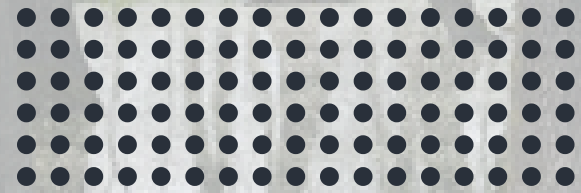
AI Flood Prediction System

Version 2.1.0

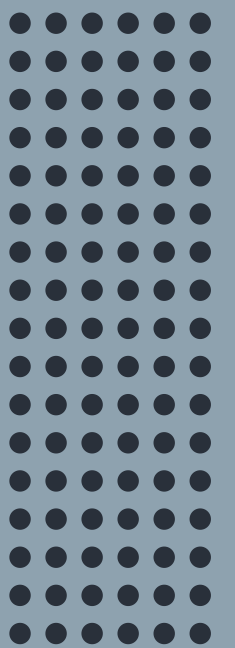
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Logout

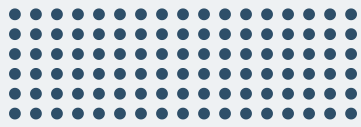
Log Out



Maintenance Team Interface



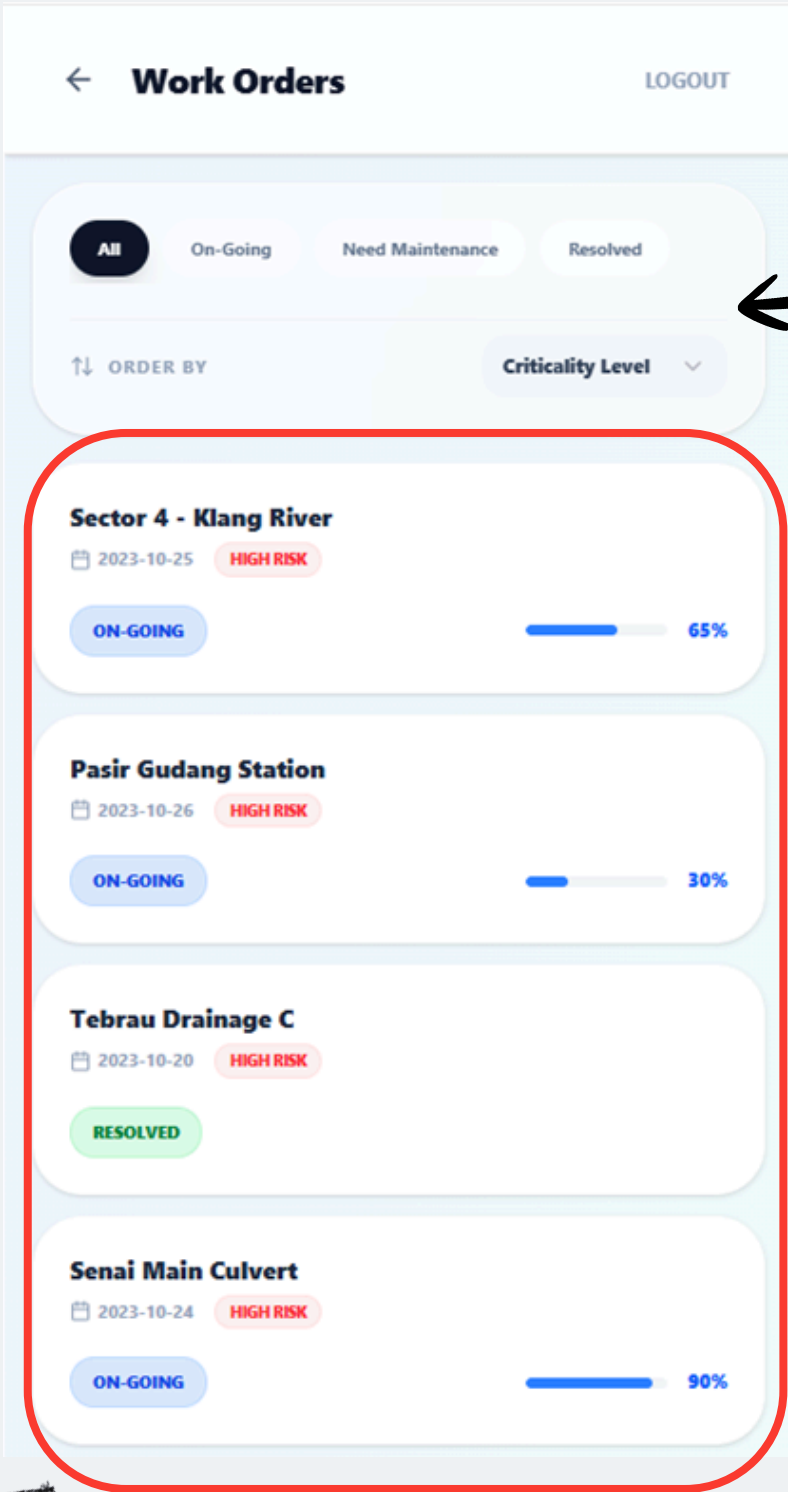
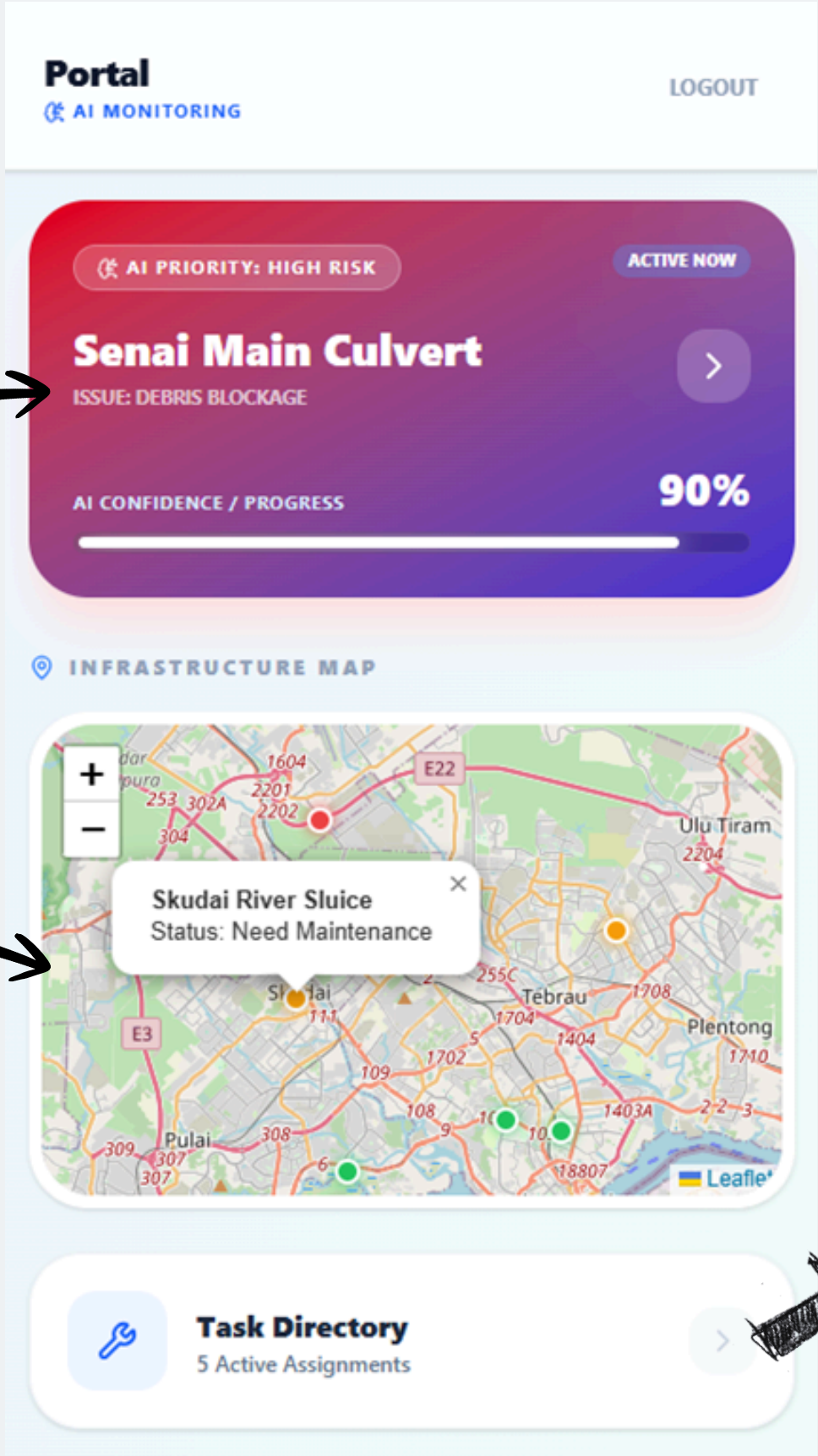
Maintenance Main Interface



**Drainage Monitoring for
Maintenance Team**

**AI Analyze Task Priority
for Maintenance Team
to Take Action**

**Real-time Mapping of
Drainage System**



Sorting With Preference

- task status
- task priority
- date

**Drainage
Maintenance Tasks
listed with Main
Information**

**Access to Work
Order Page**

Work Order and Update Log Interface

← Work Orders

LOGOUT

AllOn-GoingNeed MaintenanceResolved

ORDER BYCriticality Level

Taman Sentosa Drain

2023-10-27MEDIUM RISK

NEED MAINTENANCE

Skudai River Sluice

2023-10-28MEDIUM RISK

NEED MAINTENANCE

Kulai Catchment

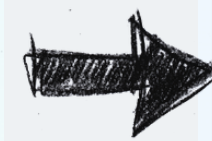
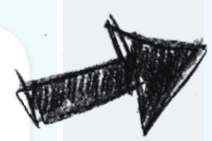
2023-10-28LOW RISK

NEED MAINTENANCE

Austin Heights Canal

2023-10-30LOW RISK

NEED MAINTENANCE



← Update Log

LOGOUT

MEDIUM RISK

REPORTED 2023-10-27

Taman Sentosa Drain

"Routine monitoring shows significant siltation in the secondary drainage canal. Recommended dredging within the next 48 hours to prevent backflow."

FIELD OPERATIONS

INITIALIZE MISSION

← Update Log

LOGOUT

MEDIUM RISK

REPORTED 2023-10-27

Taman Sentosa Drain

"Routine monitoring shows significant siltation in the secondary drainage canal. Recommended dredging within the next 48 hours to prevent backflow."

Team arrived. Initial assessment done.

FIELD OPERATIONS

LOG ACTIVITYFINALIZE

Done

Confirm Completion?

Are you sure all blockages are cleared at Taman Sentosa Drain? This will recalibrate live risk data.

Go BackYes, Task Complete

Start Maintenance

Maintenance Progress

Update Drainage Maintenance Status

Sort Task that Need Maintenance

Mission Accomplished!

Live updates sent to central HQ.

LOCATIONTaman Sentosa Drain

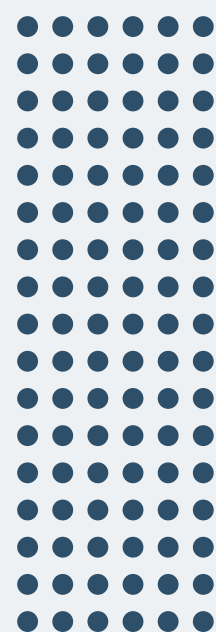
COMPLETION TIME04:31 PM Today

Flood risk levels for this sector have been recalibrated to LOW.

CLOSE REPORT

Update Status to Resolved

THANK YOU!



Group 7